From the Weight-Space Cyclic Locus to the Critical Locus

Abstract

We revisit the geometry of rank-r critical loci of deep linear networks from a direct weight-space viewpoint. In previous versions of the project, the passage from the cyclic variety to the critical locus was mediated by a residual parameter space together with a lifting map. Here we keep both objects directly inside the parameter space Ω. We define the cyclic locus by the transpose-gradient equations

$$W_{<\ell} W_0 W_{>\ell} = 0, \qquad W_0 = \Sigma_{XY} P_Q,$$

and study its rank-r part $\Gamma^r_{d,S}$. The critical locus $\mathcal{C}^r_{d,S}$ is then the distinguished fiber cut out by the additional condition $\mu(W) = P_S W^*$. The main result of this draft is that the $G_d^{W_0}$-orbits in $\Gamma^r_{d,S}$ are in bijection with the H_S-orbits in $\mathcal{C}^r_{d,S}$. This gives the direct weight-space replacement for the old lifting statement and shows formally how a stratification of the cyclic locus transfers to the critical locus.

1 Introduction

The motivation of this draft is to keep the geometry of critical points inside the original parameter space of a deep linear network. In the residual-space approach of the earlier versions, one first encoded the cyclic equations in a smaller residual variety and then lifted the corresponding data back to the critical locus by a block-normal-form construction. That point of view is effective, but it hides the geometric relation between the cyclic equations and the actual weight matrices.

The present draft works directly in weight space. The cyclic equations are written as polynomial equations in the weights, the determinantal and critical loci are defined directly in Ω, and the passage from the cyclic locus to the critical locus is expressed as an orbit correspondence rather than as a lifting construction. This shift in viewpoint is the main structural difference with `paper_v1.tex`, and it is also the main motivation for the paper.

2 Setup and Background

2.1 Notation

Notation	Meaning
L	number of weight matrices in the deep linear network
$d = (d_0, \dots, d_L)$	dimension vector
Ω	parameter space $\prod_{\ell=1}^{L} \mathrm{Mat}_{d_\ell \times d_{\ell-1}}(\mathbb{R})$
μ	product map $\mu(W) = W_L \cdots W_1$
W^*	teacher matrix $W^* = \Sigma_{YX}\Sigma_X^{-1}$
$r_{\max}$	rank of W^*
$S \subset \{1, \dots, r_{\max}\}$	subset of singular directions, with $\lvert S\rvert = r$
P_S, P_Q	complementary orthogonal projectors on $\mathbb{R}^{d_L}$
$W_0 = W_S$	closing map $\Sigma_{XY}P_Q$
Σ_d^r	rank-r determinantal locus in weight space
$\Gamma_{d,S}$	weight-space cyclic locus
$\Gamma_{d,S}^r$	rank-r cyclic locus
$\mathcal{C}_{d,S}^r$	rank-r critical locus attached to S
G_d	full base-change group $\prod_{\ell=0}^{L} \mathrm{GL}(d_\ell, \mathbb{R})$
G_{hid}	hidden-layer gauge group $\prod_{\ell=1}^{L-1} \mathrm{GL}(d_\ell, \mathbb{R})$
$G_d^{W_0}$	stabilizer of the closing map W_0 in G_d
H_S	stabilizer in $G_d^{W_0}$ of the distinguished product $P_S W^*$

2.2 Deep linear networks

Definition 2.1 (Deep linear network and product map)**.** Let

$$\Omega := \prod_{\ell=1}^{L} \mathrm{Mat}_{d_\ell \times d_{\ell-1}}(\mathbb{R}), \qquad W = (W_1, \dots, W_L) \in \Omega.$$

The associated deep linear network is the map

$$x \longmapsto W_L \cdots W_1 x,$$

and its end-to-end matrix is given by the product map

$$\mu : \Omega \to \mathrm{Mat}_{d_L \times d_0}(\mathbb{R}), \qquad \mu(W) := W_L \cdots W_1.$$

For each layer ℓ we use the partial-product notation

$$W_{>\ell} := W_L \cdots W_{\ell+1} \in \mathrm{Mat}_{d_L \times d_\ell}(\mathbb{R}), \qquad W_{<\ell} := W_{\ell-1} \cdots W_1 \in \mathrm{Mat}_{d_{\ell-1} \times d_0}(\mathbb{R}),$$

with the conventions

$$W_{>L} = I_{d_L}, \qquad W_{<1} = I_{d_0}.$$

Then

$$\mu(W) = W_{>\ell} W_\ell W_{<\ell} \qquad \text{for every } 1 \le \ell \le L.$$

Definition 2.2 (Population loss and teacher matrix)**.** The population quadratic loss is

$$\mathcal{L}(W) := \mathbb{E}\big[\|Y - \mu(W)X\|_2^2\big].$$

Writing

$$\Sigma_X = \mathbb{E}[XX^\top], \qquad \Sigma_{XY} = \mathbb{E}[XY^\top], \qquad \Sigma_{YX} = \Sigma_{XY}^\top,$$

the corresponding teacher matrix is

$$W^* := \Sigma_{YX}\Sigma_X^{-1}, \qquad r_{\max} := \operatorname{rank}(W^*).$$

Standing assumptions. Throughout this draft we assume that Σ_X is invertible, that Σ_{XY} has full rank d_L, that W^* has distinct nonzero singular values, and that the network has no bottleneck, i.e.

$$d_\ell \geq d_L \qquad \text{for every } 0 \leq \ell \leq L.$$

2.3 Gradient equations in weight space

Proposition 2.3 (Gradient formula)**.** For each $\ell \in \{1, \dots, L\}$,

$$\nabla_{W_\ell}\mathcal{L}(W) = 2\, W_{>\ell}^\top\big(\mu(W)\Sigma_X - \Sigma_{YX}\big)W_{<\ell}^\top.$$

Consequently,

$$\big(\nabla_{W_\ell}\mathcal{L}(W)\big)^\top = 2\, W_{<\ell}\big(\Sigma_X\mu(W)^\top - \Sigma_{XY}\big)W_{>\ell}.$$

Proof. The differential of the product map in the direction $\dot{W}_\ell$ is

$$D\mu(W)[\dot{W}_\ell] = W_{>\ell}\dot{W}_\ell W_{<\ell}.$$

Since the gradient of the quadratic loss with respect to the end-to-end matrix M is

$$\nabla_M\mathcal{L}(M) = 2(M\Sigma_X - \Sigma_{YX}),$$

the chain rule gives the first identity. The second follows by transposition and the symmetry of Σ_X. □

The direct weight-space cyclic equations arise by evaluating the transpose gradient on the rank-r critical sheet.

Proposition 2.4 (Weight-space cyclic equations)**.** Let W be a first-order critical point of rank r, and let $S \subset \{1, \dots, r_{\max}\}$ be the unique subset of cardinality r such that

$$\mu(W) = P_S W^*.$$

Then the transpose-gradient equations are equivalent to

$$W_{<\ell}W_0W_{>\ell} = 0, \qquad 1 \leq \ell \leq L,$$

where

$$W_0 = W_S := \Sigma_{XY}P_Q.$$

Proof. At a critical point associated with S, one has

$$\mu(W) = P_S W^* = P_S \Sigma_{YX} \Sigma_X^{-1}.$$

Hence

$$\Sigma_X \mu(W)^\top - \Sigma_{XY} = \Sigma_X \Sigma_X^{-1} \Sigma_{XY} P_S - \Sigma_{XY} = \Sigma_{XY}(P_S - I) = -\Sigma_{XY} P_Q.$$

Substituting this identity into the transpose-gradient formula yields the claimed equations. □

Definition 2.5 (Determinantal, cyclic, and critical loci)**.** Fix a rank r and a subset $S \subset \{1, \dots, r_{\max}\}$ with $|S| = r$. Define:

$$\Sigma_d^r := \{W \in \Omega : \operatorname{rank}(\mu(W)) = r\},$$

$$\Gamma_{d,S} := \{W \in \Omega \mid W_{<\ell} W_0 W_{>\ell} = 0 \text{ for every } 1 \le \ell \le L\},$$

$$\Gamma_{d,S}^r := \Sigma_d^r \cap \Gamma_{d,S},$$

and

$$\mathcal{C}_{d,S}^r := \left\{ W \in \Gamma_{d,S}^r \;\middle|\; \mu(W) = P_S W^* \right\}.$$

Remark 2.6 (Direct weight-space viewpoint)**.** This is the main departure from the residual-space setup of `paper_v1.tex`. The cyclic locus and the critical locus are both now defined directly in Ω, and the problem is to relate them without passing through a separate residual parameter space.

3 From the Cyclic Variety to the Critical Locus

3.1 Symmetry groups and the corrected orbit problem

Definition 3.1 (Relevant symmetry groups)**.** The full base-change group is

$$G_d := \prod_{\ell=0}^{L} \mathrm{GL}(d_\ell, \mathbb{R}),$$

acting on Ω by

$$(g \cdot W)_\ell := g_\ell W_\ell g_{\ell-1}^{-1}.$$

Its hidden subgroup is

$$G_{\mathrm{hid}} := \prod_{\ell=1}^{L-1} \mathrm{GL}(d_\ell, \mathbb{R}),$$

viewed as the subgroup of G_d with $g_0 = I_{d_0}$ and $g_L = I_{d_L}$. Finally, the stabilizer of the closing map is

$$G_d^{W_0} := \{g \in G_d : g_0 W_0 g_L^{-1} = W_0\}.$$

Lemma 3.2 (Equivariance of the product map)**.** For every $g \in G_d$ and every $W \in \Omega$,

$$\mu(g \cdot W) = g_L \mu(W) g_0^{-1}.$$

In particular, G_{hid} preserves $\mu(W)$ exactly.

Proof. Expanding the product and cancelling adjacent inverse factors gives

$$\mu(g \cdot W) = (g_L W_L g_{L-1}^{-1}) \cdots (g_1 W_1 g_0^{-1}) = g_L W_L \cdots W_1 g_0^{-1} = g_L \mu(W) g_0^{-1}.$$

If $g \in G_{\text{hid}}$, then $g_0 = I_{d_0}$ and $g_L = I_{d_L}$, hence

$$\mu(g \cdot W) = \mu(W).$$

□

Proposition 3.3 (Stability of the cyclic locus)**.** The loci $\Gamma_{d,S}$ and $\Gamma_{d,S}^r$ are stable under the action of $G_d^{W_0}$.

Proof. For fixed ℓ, set

$$E_\ell(W) := W_{<\ell} W_0 W_{>\ell}.$$

If $g \in G_d^{W_0}$, then

$$(g \cdot W)_{<\ell} = g_{\ell-1} W_{<\ell} g_0^{-1}, \qquad (g \cdot W)_{>\ell} = g_L W_{>\ell} g_\ell^{-1},$$

so

$$E_\ell(g \cdot W) = g_{\ell-1} W_{<\ell} (g_0^{-1} W_0 g_L) W_{>\ell} g_\ell^{-1}.$$

Since $g_0^{-1} W_0 g_L = W_0$, we obtain

$$E_\ell(g \cdot W) = g_{\ell-1} E_\ell(W) g_\ell^{-1}.$$

Thus the vanishing of all cyclic equations is preserved. Rank is also preserved because $\mu(g \cdot W) = g_L \mu(W) g_0^{-1}$. □

Remark 3.4 (Why hidden gauge is too small)**.** Because G_{hid} preserves $\mu(W)$ exactly, it cannot move a general point of $\Gamma_{d,S}^r$ into the critical locus unless the condition

$$\mu(W) = P_S W^*$$

already holds. This is why the orbit problem must be stated with a group that acts nontrivially on the endpoints.

For positive rank, the critical locus is not stable under the full group $G_d^{W_0}$.

Proposition 3.5 (Non-stability of $\mathcal{C}_{d,S}^r$ for $r > 0$)**.** Assume $r > 0$. Then $\mathcal{C}_{d,S}^r$ is not stable under the action of $G_d^{W_0}$.

Proof. Choose $\lambda \in \mathbb{R}^\times$ with $\lambda \neq 1$, and define

$$g_0 = I_{d_0}, \qquad g_L = \lambda P_S + P_Q, \qquad g_\ell = I_{d_\ell} \ \text{ for } 1 \leq \ell \leq L-1.$$

Then g_L is invertible with inverse $\lambda^{-1} P_S + P_Q$, and

$$g_0 W_0 g_L^{-1} = \Sigma_{XY} P_Q (\lambda^{-1} P_S + P_Q) = \Sigma_{XY} P_Q = W_0.$$

So $g \in G_d^{W_0}$. If $W \in \mathcal{C}_{d,S}^r$, then

$$\mu(g \cdot W) = g_L (P_S W^*) = \lambda P_S W^*.$$

Since $r > 0$, the matrix $P_S W^*$ is nonzero, so

$$\mu(g \cdot W) \neq P_S W^*.$$

Hence $g \cdot W \notin \mathcal{C}_{d,S}^r$. □

Definition 3.6 (Correct stabilizer on the critical locus)**.** Define

$$H_S := \mathrm{Stab}_{G_d^{W_0}}(P_S W^*) = \left\{ g \in G_d^{W_0} \;\middle|\; g_L (P_S W^*) g_0^{-1} = P_S W^* \right\}.$$

Remark 3.7 (Correct orbit spaces)**.** For $r > 0$, the relevant orbit spaces are

$$\Gamma_{d,S}^r / G_d^{W_0} \qquad \text{and} \qquad \mathcal{C}_{d,S}^r / H_S.$$

The first direct weight-space theorem is precisely the statement that these two orbit spaces are in bijection.

3.2 Endpoint reduction

Definition 3.8 (Endpoint group)**.** Define the endpoint group

$$K_S := \{(g_0, g_L) \in \mathrm{GL}(d_0, \mathbb{R}) \times \mathrm{GL}(d_L, \mathbb{R}) : g_0 W_0 g_L^{-1} = W_0\}.$$

It acts on the function space $\mathrm{Mat}_{d_L \times d_0}(\mathbb{R})$ by

$$(g_0, g_L) \cdot M := g_L M g_0^{-1}.$$

Lemma 3.9 (Projection to the endpoint group)**.** The map

$$\pi : G_d^{W_0} \to K_S, \qquad \pi(g_0, \ldots, g_L) := (g_0, g_L),$$

is a surjective homomorphism, and for every $W \in \Omega$ one has

$$\mu(G_d^{W_0} \cdot W) = K_S \cdot \mu(W).$$

Proof. The projection is well-defined because $g_0 W_0 g_L^{-1} = W_0$ is exactly the defining relation of K_S. It is surjective because for every $(g_0, g_L) \in K_S$, the element

$$\widetilde{g} = (g_0, I_{d_1}, \ldots, I_{d_{L-1}}, g_L)$$

lies in $G_d^{W_0}$ and projects to (g_0, g_L). The orbit identity follows at once from equivariance of μ. □

Corollary 3.10 (Endpoint reduction)**.** For $W \in \Gamma_{d,S}^r$, the following are equivalent:

1. there exists $g \in G_d^{W_0}$ such that $g \cdot W \in \mathcal{C}_{d,S}^r$;
2. there exists $(g_0, g_L) \in K_S$ such that
$$g_L \mu(W) g_0^{-1} = P_S W^*;$$
3. the product $\mu(W)$ lies in the K_S-orbit of $P_S W^*$.

Proof. The implication (1)⇒(2) is just equivariance of μ. The implication (2)⇒(1) follows by lifting (g_0, g_L) to an element of $G_d^{W_0}$ and using stability of $\Gamma_{d,S}^r$. The equivalence between (2) and (3) is the statement that two points lie in the same orbit if and only if one is obtained from the other by the group action. □

3.3 Visible invariants and normalization

Proposition 3.11 (Visible product-level constraints)**.** If $W \in \Gamma_{d,S}$ and $M = \mu(W)$, then

$$W_0 M = 0 \qquad \text{and} \qquad M W_0 = 0.$$

Consequently,

$$\mathrm{im}(M) \subseteq \mathrm{Ker}(W_0) \qquad \text{and} \qquad \mathrm{im}(W_0) \subseteq \mathrm{Ker}(M).$$

Proof. For $\ell = 1$, the cyclic equation reads

$$W_0 W_{>1} = 0.$$

Multiplying on the right by W_1 gives

$$W_0 M = 0.$$

For $\ell = L$, the cyclic equation reads

$$W_{<L} W_0 = 0.$$

Multiplying on the left by W_L gives

$$M W_0 = 0.$$

The image-kernel statements are immediate. □

Corollary 3.12 (Rank-r sharpening)**.** Assume that Σ_{XY} has full rank d_L. If $W \in \Gamma^r_{d,S}$ and $M = \mu(W)$, then

$$\mathrm{Ker}(W_0) = \mathrm{im}(P_S), \qquad \mathrm{rank}(W_0) = d_L - r, \qquad \mathrm{im}(M) = \mathrm{im}(P_S).$$

Proof. Because Σ_{XY} has full rank d_L, it is injective as a map from $\mathbb{R}^{d_L}$ to $\mathbb{R}^{d_0}$. Hence

$$\mathrm{Ker}(W_0) = \mathrm{Ker}(\Sigma_{XY} P_Q) = \mathrm{Ker}(P_Q) = \mathrm{im}(P_S).$$

Since M has rank r and $\mathrm{im}(M) \subseteq \mathrm{Ker}(W_0)$, the inclusion is an equality. □

Theorem 3.13 (Normalization theorem for the end-to-end map)**.** Assume that Σ_{XY} has full rank d_L, and that the distinguished critical product $P_S W^*$ occurs for the subset S. Then for every

$$W \in \Gamma^r_{d,S},$$

one has

$$\mu(W) \in K_S \cdot (P_S W^*).$$

Proof. Set

$$\sigma := W_0 = \Sigma_{XY} P_Q, \qquad T := P_S W^*, \qquad U := \mathrm{Ker}(\sigma), \qquad I := \mathrm{im}(\sigma).$$

For any matrix M satisfying

$$\sigma M = 0 \qquad \text{and} \qquad M \sigma = 0,$$

the map M factors through a reduced map

$$\overline{M} : \mathbb{R}^{d_0}/I \to U.$$

Moreover, the group K_S acts on reduced maps through the natural left-right action of

$$\mathrm{GL}(U) \times \mathrm{GL}(\mathbb{R}^{d_0}/I),$$

and this action is surjective because one can choose complements of U and I and build endpoint transformations blockwise.

Now let $M = \mu(W)$ with $W \in \Gamma^r_{d,S}$. The previous proposition gives

$$\sigma M = 0 \qquad \text{and} \qquad M\sigma = 0,$$

and the rank-r sharpening shows that $\overline{M}$ is surjective. The model critical product $T = P_S W^*$ satisfies the same two annihilation relations, has rank r, and has image U. Therefore its reduced map

$$\overline{T} : \mathbb{R}^{d_0}/I \to U$$

is also surjective.

Since any two surjective linear maps from $\mathbb{R}^{d_0}/I$ to U are equivalent under left-right multiplication by invertible maps, there exists $(g_0, g_L) \in K_S$ with

$$g_L T g_0^{-1} = M.$$

Hence

$$M \in K_S \cdot T = K_S \cdot (P_S W^*),$$

as required. □

3.4 Orbit correspondence

Theorem 3.14 (First direct weight-space theorem: orbit-surjectivity)**.** Assume that Σ_{XY} has full rank d_L. Then every $G_d^{W_0}$-orbit in $\Gamma^r_{d,S}$ meets the critical locus $\mathcal{C}^r_{d,S}$.

Proof. Let $W \in \Gamma^r_{d,S}$. By Theorem 3.13,

$$\mu(W) \in K_S \cdot (P_S W^*).$$

By endpoint reduction, there exists $g \in G_d^{W_0}$ such that

$$g \cdot W \in \mathcal{C}^r_{d,S}.$$

□

Theorem 3.15 (Second direct weight-space theorem: orbit-bijection)**.** Assume that $r > 0$ and that Σ_{XY} has full rank d_L. Then every $G_d^{W_0}$-orbit in $\Gamma^r_{d,S}$ meets $\mathcal{C}^r_{d,S}$ in exactly one H_S-orbit. Equivalently, the inclusion induces a bijection

$$\mathcal{C}^r_{d,S}/H_S \xrightarrow{\sim} \Gamma^r_{d,S}/G_d^{W_0}.$$

Proof. Existence is Theorem 3.14. For uniqueness, let $W, W' \in \mathcal{C}^r_{d,S}$ lie in the same $G_d^{W_0}$-orbit. Then $W' = g \cdot W$ for some $g \in G_d^{W_0}$. Since both points lie in the critical locus,

$$\mu(W) = P_S W^* \qquad \text{and} \qquad \mu(W') = P_S W^*.$$

Equivariance gives

$$P_S W^* = \mu(W') = g_L \mu(W) g_0^{-1} = g_L (P_S W^*) g_0^{-1}.$$

Hence $g \in H_S$. This proves that

$$\left(G_d^{W_0} \cdot W\right) \cap \mathcal{C}_{d,S}^r = H_S \cdot W.$$

The bijection of orbit sets follows formally from existence and uniqueness. □

Corollary 3.16 (Transfer of stratifications)**.** Suppose that the rank-r cyclic locus admits a decomposition into locally closed $G_d^{W_0}$-stable strata

$$\Gamma_{d,S}^r = \bigsqcup_{\alpha \in A} \mathcal{O}_\alpha,$$

where each stratum is a union of $G_d^{W_0}$-orbits. Then the critical locus inherits the decomposition

$$\mathcal{C}_{d,S}^r = \bigsqcup_{\alpha \in A} \mathcal{C}_\alpha, \qquad \mathcal{C}_\alpha := \mathcal{C}_{d,S}^r \cap \mathcal{O}_\alpha,$$

and this is a stratification by locally closed H_S-stable pieces.

Proof. By Theorem 3.15, every ambient $G_d^{W_0}$-orbit meets the critical locus in exactly one H_S-orbit. Therefore the orbit partition of the cyclic locus induces an H_S-orbit partition of the critical locus. Grouping those critical orbits according to the ambient strata $\mathcal{O}_\alpha$ gives the claimed decomposition. □

Remark 3.17 (Role of the first theorem in the paper)**.** The theorem above is the direct weight-space replacement of the old lifting statement. It does not identify the cyclic locus with the critical locus. Instead, it identifies their orbit spaces in the precise way needed to transfer combinatorial data, such as cyclic rank patterns, from $\Gamma_{d,S}^r$ to $\mathcal{C}_{d,S}^r$.

4 Stratification of Cyclic Quivers with Finite Relation

This section will explain how to stratify the cyclic locus $\Gamma_{d,S}^r$ itself by orbit-type data, and in particular by cyclic rank patterns or multiplicity data coming from the corresponding quiver representations.

5 Application: Stratification of the Critical Loci

Combining the orbit correspondence of Theorem 3.15 with a classification of $G_d^{W_0}$-orbits in $\Gamma_{d,S}^r$ will produce the desired stratification of the critical locus $\mathcal{C}_{d,S}^r$.

6 Discussion and Future Work

The direct weight-space formalism developed here is intended to make the critical-locus geometry more visible from the original parameters of the deep linear network. The next steps are to classify the cyclic orbits themselves and to connect the resulting critical strata with the saddle-influence questions driving the larger project.